CYBER THREAT INTELLIGENCE

25CS2263

STUDENT ID:

STUDENT NAME:

ACADEMIC YEAR: 2024-25

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A.Y. 2024-25 LAB/SKILL CONTINUOUS EVALUATION

S.No	Date	Experiment Name	Pre-Lab (10M)	In-Lab (25M)			Post-Lab (10M)	Viva Voce (5M)	Total (50M)	Faculty Signature
				Program/ Procedure (5M)	Data and Results (10M)	Analysis & Inference (10M)				
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1. CYBER THREAT INTELLIGENCE LIFE CYCLE PHASES

Aim/Objective: To provide foundational knowledge of Cyber Threat Intelligence and introduce life cycle, enabling learners to understand how data is transformed into valuable insightsthrough structured processes and tools.

Pre-Lab:

1. What is use of the prefix word CYBER

2. What is Threat and its different type

3. Define Threat intelligence

4. Difference between Threat and Vulnerability

5. Give example for Cyber threat Intelligence

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In-Lab:
Implement a small fictional company scenario, identify 5–7 intelligence requirements and classify outputs as strategic, operational, tactical, or technical; then map each to the CTI lifecycle phases

- **Procedure/Program:**

- **Data and Results:**

- **Analysis and Inferences:**

VIVA-VOCE Questions (In-Lab):

1. **What is** strategic intelligence

2. **Define** operational intelligence

3. **Give example for** tactical intelligence

4. **What is** technical intelligence

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Post-Lab: Develop a CTI life cycle for an organisation -KL University

- **Procedure/Program:**

- **Data and Results:**

- **Analysis and Inferences:**

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2. Implementation a Diamond Model (adversary, victim, infrastructure, capability)for cerber Kill Chain

Aim/Objective: To provide foundational knowledge Identify and label the four Diamond Model vertices (adversary, victim, infrastructure, capability) for at least one intrusion scenario across all Cyber Kill Chain phases

Pre-Lab:

1. Draw or document one or more diamonds per kill-chain phase, showing how changes in infrastructure reflect activity

2. Derive at least three actionable intelligence outputs from the model (e.g., candidate detection rules, pivot points for hunting, or hypotheses

3. Explain how relationships between vertices (e.g., adversary–infrastructure, infrastructure–victim) help prioritize collection, enrichment in CTI

IN-Lab

Implementation a Diamond Model (adversary, victim, infrastructure, capability)forCyber Kill

Chain

(a)identify which events belong to each Cyber Kill Chain phase: Recon, Weaponization,

Delivery, Exploitation, Installation, Actions on Objectives

(b) A table with one or more observable events per phase and the evidence that supports their placement

- **Procedure/Program:**

- **Data and Results:**

- **Analysis and Inferences:**

VIVA-VOCE Questions (In-Lab):

1. Define Cyber Kill Chain

2. Give example for Recon, Weaponization, Delivery

3. Explain Actions on Objectives

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Post Lab:

1. How did combining Cyber Kill Chain with the Diamond Model change the way?

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3.Implementation to Select one APT group

AIM
To analyse a real-world nation-state threat actor using CTI methodologies.

Pre Lab

1. **What is an APT group? How is it different from a cybercriminal gang?**

2. **What is the Intelligence Lifecycle and how does it apply to CTI?**

3. **Why is attribution difficult in nation-state cyber operations?**

4. **What information sources are used to build a threat actor profile?**

5. **What are common motivations behind state-sponsored threat actors?**

In lab
Implementation to Select one APT group and create a 2–3-page profile: history, typical targets, motivations-

- **Procedure/Program:**

- **Data and Results:**

- **Analysis and Inferences:**

VIVA-VOCE Questions (In-Lab):

1. Which APT group did you select and why?

2. How did you handle multiple names or aliases for the same APT group

Post Lab
Add a one-page appendix mapping the group's key TTPs to MITRE ATT&CK and highlighting 3–5 highest priority Technique

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4. Implement OSINT Tools for Cybersecurity

Aim : To conduct a scoped OSINT investigation against a fictitious or pre-approved target, organize collected data, derive security-relevant insights, and produce a short intelligence report that could support cybersecurity decision-making

Pre Lab :

1. Definition of OSINT and its role in cybersecurity

2. Explain OSINT Tools

   TheHarvester,

   SpiderFoot

   Recon-ng

   Shodan

   OSINT Framework

   FOCA (metadata)

3. Convert cleaned data into intelligence by the following

Identifying patterns of behavior:

Mapping infrastructure:

Profiling threat actors:

Recognizing vulnerabilities:

Assessing credibility of sources:

In lab

Implement OSINT Tools for Cybersecurity

- **Procedure/Program:**

- **Data and Results:**

- **Analysis and Inferences:**

VIVA-VOCE Questions (In-Lab):

1. What is OSINT, and how is it different from general web browsing or traditional intelligence collection?

2.Why are legal and ethical considerations critical in OSINT, and what activities are clearly not allowed in OSINT

3.Give examples of public sources hat qualify as OSINT in cybersecurity

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Post Lab

1.Compare two tools (e.g., Shodan vs. theHarvester).

2.Write a short paragraph evaluating the overall digital footprint of the target.

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5.Using a fictional domain to perform OSINT: WHOIS, DNS records, Shodan exposure
Aim: To perform Open-Source Intelligence (OSINT) on a **fictional domain**, and gather intelligence using **WHOIS lookup, DNS record enumeration, and Shodan exposure analysis**, while following legal and ethical passive reconnaissance practices.
In lab:

1. What type of information does WHOIS provide about a domain?

2. Why did the fictional domain return no DNS records?

3. What type of data can Shodan reveal for real-world domains or Ips?

4. Explain the security risks associated with exposed ports and outdated technologies.

5. How does passive reconnaissance differ from active scanning?

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In lab

In fictional domain to perform OSINT: WHOIS, DNS records, Shodan exposure

- **Procedure/Program:**

- **Data and Results:**

- **Analysis and Inferences**

VIVA Questions

1.**What WHOIS information helps an attacker?**

2. Write Risky DNS records?

3.. List out OSINT limitations

Post Lab

Link between WHOIS + DNS + Shodan as a unified intelligence picture

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6. Design framework for basic Maltego/Reconn-ng/SpiderFoot scan and GitHub/social footprint search

Aim : To perform an end-to-end OSINT investigation on a fictional entity using Maltego, Recon-ng, and SpiderFoot, and to analyze GitHub activity and social media footprint to identify digital exposure, relationships, associated assets, and potential risks.

Pre lab :

1.What OSINT is and why passive reconnaissance matters

2.Purpose of graph-based OSINT (Maltego)

3.Purpose of automated enumeration (SpiderFoot)

4.Purpose of modular CLI-based framework (Recon-ng)

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In lab

(a) Maltego Basic Scan-Build a relationship graph of a persona/domain
(b) Recon-ng Workspace Scan-Perform modular OSINT enumeration
(c) SpiderFoot Automated Scan, GitHub Footprint Investigation

Procedure/Program:

Data and Results

Analysis and Inferences

Viva Qustions

1. How can Maltego, Recon-ng, and SpiderFoot complement each other?

2. Give an example of how GitHub leaks can lead to exploitation.

3. How could machine learning improve OSINT automation?

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Post Lab

1.Which OSINT tool produced the most valuable results?

2. Did any tool generate false positives? How were they handled?

3. How does correlation improve intelligence accuracy?

4. What risks were identified in the fictional domain?

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7.Examine ,In small groups, research the SolarWinds or Colonial Pipeline incident and reconstruct: kill chain, ATT&CK techniques, indicators, initial access vector, and attribution challenge.

Aim:

To analyse a major real-world cyber incident (SolarWinds OR Colonial Pipeline) and reconstruct its attack lifecycle using: Cyber Kill Chain, MITRE ATT&CK technique, Indicators of Compromise (IOCs)initial Access Vector

Pre Lab

1. Explain the difference between **TTPs**, **IOCs**, and **attack vectors**.

2. What makes attribution difficult in nation-state–level intrusions?

3. Identify three common initial access techniques used by APTs.

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In lab :

SolarWinds or Colonial Pipeline incident and reconstruct: kill chain, ATT&CK techniques, indicators, initial access vector, and attribution challenge.

Procedure/Program:

Data and Results

Analysis and Inferences

VIVA QUESTION

1. What is the cyber kill chain and why is it useful?

2. How does ATT&CK differ from the kill chain?

3. Define initial access and give examples.

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Post Lab

1.How was the SolarWinds update weaponized?

2.What is Golden SAML?

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8.Implement From 2–3 different public threat reports on the same malware or campaign, extract IoCs, normalize them, remove duplicates, and validate with one open source enrichment service (e.g., VirusTotal lookup or similar).

Aim:

To understand and implement the process of collecting Indicators of Compromise (IoCs) from multiple public threat intelligence reports, normalize and deduplicate them, and validate the IoCs using an open-source enrichment service such as VirusTotal, AlienVault OTX, AbuseIPDB, or URLScan.

Pre lab

What Are IoCs?

IoCs are forensic artifacts that indicate malicious activity. Explain :

- File hashes (MD5/SHA1/SHA256)
- IP addresses (C2 servers)
- Domains/Subdomains
- URLs
- Registry keys
- Mutexes
- Filenames
- Email addresses / phishing lures

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In lab Implement From 2–3 different public threat reports on the same malware or campaign, extract IoCs, normalize them, remove duplicates, and validate with one open source enrichment service (e.g., VirusTotal lookup or similar).

Procedure/Program:

Data and Results

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Analysis and Inferences

VIVA QUESTION

1. What is an IoC? And its type

2. What are limitations of using VirusTotal?

3. How does multi-source IoC validation aid threat hunting?

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Post Lab

1.If two vendors disagree on maliciousness, how will you handle it?

2.What is the difference between static and dynamic IoCs?

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9.Implement the log sources needed, correlation logic, and alert fields enriched with CTI context (threat family, actor, geo, confidence).

Aim To implement and configure security monitoring using multiple log sources, create correlation logic for threat detection, and enrich alerts with Cyber Threat Intelligence (CTI) context including threat family, actor, geo-location, and confidence level.

Pre Lab

1.Understand mapping of logs to MITRE ATT&CK techniques.

2.Review CTI feed fields: threat family, actor, campaign, TTP, geolocation.

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In Lab

Implement the log sources needed, correlation logic, and alert fields enriched with CTI context (threat family, actor, geo, confidence).-Configure Log Sources, Implement Correlation Logic, Enrich Alerts with CTI, Validate Alerts

Procedure/Program:

Data and Results

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Analysis and Inferences

VIVA QUESTION

1.What fields are mandatory in a CTI-enriched alert?

2.Give an example of a correlation rule mapping to MITRE ATT&CK.

3.How do you determine the confidence level of an alert?

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Post Lab

Develop a CTI-Enriched Alert Template

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10. Analyze Based on an ATT&CK technique (e.g., phishing or lateral movement), write:
A CTI note summarizing adversary behaviour and relevant IoCs,
3–5 specific hunting hypotheses and corresponding log queries/events

Aim:

To analyze adversary behaviour associated with **phishing (MITRE ATT&CK T1566)**, identify **relevant IoCs**, and develop **3–5 threat hunting hypotheses** along with **log queries/events** for detection in an enterprise environment.

Pre Lab

1. What is Phishing

2. Why attackers use phishing

3. Expected Data Sources

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In lab

Analyze Based on an ATT&CK technique (e.g., phishing or lateral movement), write:
A CTI note summarizing adversary behaviour and relevant IoCs,
3–5 specific hunting hypotheses and corresponding log queries/events
Procedure/Program:

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Data and Results

Analysis and Inferences

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VIVA Question

1. Difference between IoC and IoA?

2. Which logs help detect phishing attacks?

3. What is a hunting hypothesis?

Post Lab

1.Why are macros often used in phishing payloads?justify

2.What are signs of suspicious authentication behaviour?

3.Why is hypothesis-driven hunting effective in CTI?

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11. Implement chosen malware family or technique from previous practical
One basic YARA rule (file or string based), One Sigma rule for a Windows or Linux log event related to that behaviour

Aim

To map adversary behaviour to actual detection artifacts.

Pre lab

1. Explain Basic YARA syntax:- Strings, conditions, file scanning

2. Sigma rule format:

   - YAML structure
   - Logsource, detection, condition

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In lab

Implement chosen malware family or technique from previous practical
One basic YARA rule (file or string based), One Sigma rule for a Windows or Linux log event related to that behaviour
Procedure/Program:

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Data and Results

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Post lab

1.Use the YARA rule on mail gateways or file upload scanners.

2.Deploy the Sigma rule to SIEM to alert SOC analysts

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12. Create a case study by using Tag IoCs with threat actor / campaign / TLP, STIX/TAXI

Aim To analyze a cyber threat campaign, extract and tag Indicators of Compromise (IoCs) with relevant threat actor/campaign associations and TLP markings, and represent the findings using STIX 2.1 objects and TAXII-compliant sharing format.

Pre lab

1. Explain Threat actor profiling basics

2. Explain Indicator, Malware Threat Actor, Campaign

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In lab: Create a case study by using Tag IoCs with threat actor / campaign / TLP, STIX/TAXI

Procedure/Program:

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Data and Results

Analysis and Inferences

VIVA Question

1. What is the purpose of TLP?

2. Name three STIX objects you used in this lab.

3. What is TAXII used for?

4. How do ATT&CK techniques fit into threat campaigns?

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Post Lab

1.IoCs extracted and tagged with threat actor + campaign

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